Supplementary Information

For the manuscript "Context-Dependent Adaptation in Structured Environments" by Yuya Karita, Gisela T. Rodríguez-Sánchez, Elisa Brambilla, J. Carlos R. Hernandez-Beltran, Michael Schwarz, and Paul B. Rainey, published from Proceedings of the Royal Society B [DOI: 10.1098/rspb.2025.2004].

Supplementary Methods

Confocal imaging and 3D reconstruction of microcolonies.

To observe the 3D structure of an ALI microcolony, a *wspF* WS mutant with a fluorescent tag (sGFP2) was cultured in a 35 mm plastic dish (Sarstedt) and observed with an upright confocal microscope (LSM880, Zeiss). To reduce the autofluorescence, lysogeny broth (LB) was used instead of KB. Moreover, *pvdS*, a gene necessary for expression of the fluorescent siderophore pyoverdin, was deleted in order to eliminate the pyoverdin-generated fluorescence signal. Static culture was started from a low inoculum ($\sim 10^4$ cells/mL) and observed after 13 hours of static incubation. Colonies at the ALI were captured with a 10X objective, closely located above the ALI. To capture rafting colonies, the speed of image acquisition was increased to minimize drift, albeit at the expense of image quality. The z-stacks of the confocal images were reconstructed by the 3D viewer of ImageJ.

Co-culture experiments to capture the spatial configuration of SM and WS.

To capture the interactions between SM and WS at a micron scale, *pvdS* was deleted from bacteria, and LB, instead of KB, was used as the culture medium to reduce the autofluorescence. SM and WS (*wspF* mutant) carrying fluorescent markers (sGFP2 and mScarlet) were co-cultured for 16 hours under static conditions from a low inoculum ($\sim 10^3$ cells/mL). The spatial configuration of the two types was captured with an upright microscope (Axio Imager Z2, Zeiss).

Supplementary Texts

A Stability condition for floating microcolonies at an ALI

Things smaller than the capillary length of liquid can float at the air-liquid interface even if they are heavier than the liquid. Two lifting forces, buoyancy and surface tension, counter a gravitational fall. The gravitational force can be expressed by the mass density of a colony ρ_c, the volume of the colony V, and the gravitational acceleration g,

$$\rho_c V_c g.$$

The buoyancy term is proportional to the mass of the displaced liquid by the colony. Letting the displaced liquid volume be V_{below}, the buoyant force is

$$\rho_l V_{below} g,$$

where ρ_l is the mass density of the liquid. Here, we assume $\rho_l < \rho_c$ and $V_{below} < V_c$. The lifting force by surface tension works at the air-liquid-colony interface, and can be calculated by a line integral across the periphery:

$$\int_\partial dl \gamma \sin\theta_l,$$

where $\int_\partial dl$ is an integral over the periphery, γ is the surface tension of the liquid, and θ_l is the tilting angle of the deformed liquid surface near the colony. In general, θ_l is very small and sensitive to the local properties, such as the shape or the hydrophobicity of the colony. Here, for simplicity, we assume a constant θ across the periphery and get

$$\gamma \sin\theta L_c,$$

where L_c is the colony periphery length. The force balance equation of a floating colony is

$$\rho_c V_c g = \rho_l V_{below} g + \gamma \sin\theta L_c. \tag{1}$$

For simplicity, we assumed that V_{below} was proportional to V_c:

$$V_{below} = a V_c,$$

where $a(<1)$ is a constant. The equation 1 can be simplified as

$$\left(1-\frac{a\rho_l}{\rho_c}\right)\rho_c V_c g = \gamma \sin\theta L \Leftrightarrow M_c = \frac{\gamma\sin\theta}{g}\left(1-\frac{a\rho_l}{\rho_c}\right)^{-1} L_c,$$

where $M_c = \rho_c V_c$ is the total biomass of the colony. By defining

$$G(\theta) = \frac{\gamma\sin\theta}{g}\left(1-\frac{a\rho_l}{\rho_c}\right)^{-1},$$

we get the simplified force balance equation

$$M_c = G(\theta)L_c. \tag{2}$$

As a colony gets bigger, the left-hand side grows as $\sim O(L^3)$, while the right-hand side grows as $\sim O(L^1)$. Therefore, when the colony gets too big, no θ can satisfy the equation 2, and the colony falls down from the air-liquid interface.

In computational simulations, we fixed θ regardless of the colony shape and size, and defined a constant G to evaluate the stability of a colony.

$$M_c \begin{cases} \leq GL_c & \text{float} \\ > GL_c & \text{collapse} \end{cases} \tag{3}$$

We empirically chose G to let an isolated colony collapse at the colony size consistent with the experimental result (Fig. 1).

B Simulations of early-stage spatial dynamics of ALI colonization

Static microbial culture is modeled as a combination of two populations, the well-mixed population in the bulk liquid phase (supposedly mixed by hydrodynamic convection, often driven by biological activities[1,2]) and the spatial ALI population on a two-dimensional lattice. The population density in the liquid phase $c(t)$ is modeled as a logistic growth with the growth rate r_l and the carrying capacity c_{max}.

$$\frac{dc(t)}{dt} = r_l c\left(1-c/c_{max}\right).$$

Note that r_l should be smaller than that of a shaken culture because of oxygen depletion. The starting cell density c_0 is varied in the simulation to investigate inoculum effects. From the liquid phase, cells migrate to the air-liquid interface and colonize a random position at migration flux μ, which is proportional to $c(t)$:

$$\mu = C_\mu c(t) L_{box}^2,$$

where C_μ is a coefficient, and L_{box}^2 is the entire area of the air-liquid interface.

The ALI population is modeled on a two-dimensional discrete space with a periodic boundary condition. Once a cell occupies a site, the cellular colony grows in the horizontal (spatial expansion) and vertical (local biomass accumulation) directions. The horizontal expansion follows the Eden-model-type surface growth[3]: only the cells facing to an empty space can expand and colonize a neighbor site.

To model the local biomass accumulation, we referred to the three-dimensional structure of a colony on a plate. It has been known that the height of a colony saturates due to the limited nutrient diffusion[4], and the saturation can be modeled by a logistic growth[5]. Thus, the biomass at a site (x,y), $M(x,y,t)$, follows

$$\frac{dM(x,y,t)}{dt} = rM\left(1-\frac{M}{K}\right),$$

where r is the vertical growth rate and K is the maximum local biomass. Note that intra-colony displacements of biomass, such as the diffusion term $D\nabla^2 M$, are not incorporated into the model for simplicity, and thus the biomass grows only locally. During the simulation, the stability of colonies (eq. 3) is tested at each time point, and unstable colonies are removed.

We also implemented the motility of cells as a stochastic replacement from a colony periphery to another random site. The replacement probability is defined by the rate r_s. We set $r_s = 0$ for non-motile simulations.

To conduct simulations, the modified next reaction method[6,7] is used. Note that the well-known Gillespie algorithm is not simply applicable to our model because we have the time-dependent variable $c(t)$, which defines the propensity of migration from the bulk liquid to the air-liquid interface. Fortunately, the logistic function $c(t)$ enables us to analytically treat the propensity in the method. In short, the logistically growing population

$$c(t) = \frac{c_{max}}{1+e^{-r_l t}\left(\frac{c_{max}}{c_0}-1\right)}$$

defines the propensity of migration, and the expected time to the next event Δt can be calculated by

$$\int_t^{t+\Delta t} B\frac{r_l}{c_{max}}c(t')dt' = \alpha \Leftrightarrow \Delta t = \frac{1}{r_l}\ln\left[(e^{r_l t}+A)e^{\frac{\alpha}{B}}-A\right]-t,$$

where A and B are constant, and α is related to the "internal time" of the next reaction method. Refer to[7] for further details of the method. The step-by-step description of the model simulations is as follows:

Detailed algorithm

The model has three stochastic processes: Eden-model-type colony expansion (g), migration from the bulk liquid to the ALI (m), and cell replacement due to swimming motility (s). We label the processes by index k that can take $\{g,m,s\}$. The propensity of each process $a_k(t)$ can be expressed as

$$a_g = r_g N_p,$$
$$a_m = C_\mu c(t) L_{box}^2,$$
$$a_s = r_s N_p,$$

where N_p is the total number of periphery sites. We set the timescale of the simulation to let $r_g = 1$. We define the internal time of each stochastic process T_k and the expected timing of the process in the internal time P_k. With this setting, the expected timing of the next process in the real time Δt_k is a solution of

$$\int_t^{t+\Delta t_k} a_k(t')dt' = P_k - T_k.$$

Note that for $k \in \{g,s\}$, a_k is constant given no process happens during Δt_k, and the equation above can be easily solved. For $k = m$, fortunately, a logistic function allows analytic treatments, and no numerical analysis is required. The step-by-step algorithm of the simulation is:

1. Initialize the system. Set the initial population size of the bulk liquid and the ALI. Set $t = 0$ and $P_k = T_k = 0$.
2. Generate independent random numbers s_k from a uniform distribution over $(0,1)$. Set $P_k = \ln\frac{1}{s_k}$.
3. Calculate Δt_k as the solution of $\int_t^{t+\Delta t_k} a_k(t')dt' = P_k - T_k$.
4. Set the actual time increment $\Delta = \min_k\{\Delta t_k\}$. If $\Delta \le T_{mass}$, let k' be the process giving the minimum Δt_k. If $\Delta > T_{mass}$, set $\Delta = T_{mass}$, and no process happens except for the biomass growth ($k' = \phi$).
5. Set $T_k \leftarrow T_k + \int_t^{t+\Delta} a_k(t')dt'$, and then set $t \leftarrow t+\Delta$.
6. Update the biomass of each site. And then update the site occupations based on the process k' (if $k' \neq \phi$).
   - For $k' = g$: Choose a random periphery site, and occupy one empty neighbor site.
   - For $k' = m$: Occupy a random site. If the site is already occupied, the migration fails.
   - For $k' = s$: Choose a random periphery site (x,y). Draw a random number d from a Gaussian distribution with zero mean and variance σ^2. Scan the $3\sigma \times 3\sigma$ neighbors of (x,y) to find an empty site at a distance closest to $|d|$. If there are multiple candidates, choose one randomly. Colonize the new site with biomass m_0. A swimming event fails if $d < 0.5$. We referred to the scheme in the previous literature[8] but modified it to increase the computational efficiency.
7. Check the stability of colonies. Remove unstable colonies.
8. For k', generate a new random number $s_{k'}$ from a uniform distribution over $(0,1)$, and set $P_{k'} \leftarrow P_{k'} + \ln\frac{1}{s_{k'}}$.
9. Iterate the steps 3-8 until the terminating condition is reached,

Note that this model was designed for conceptualizing the early-stage dynamics of mat formation to investigate the inoculum effect. To form a robust mat structure, strong adhesion to the side walls or mechanical cell-cell inter-connections by amyloid fibers should be required. We did not incorporate these additional complexities into our model.

C Simulations of dispersal-driven evolution.

To simulate evolutionary dynamics in an unshaken culture, we propose a simple four-state model. The model has 4 variables S_A, S_B, W_A, W_B, representing the population sizes of two cell types (SM and WS) and two spatial environments (bulk liquid and ALI). The simplest equations for the time development of the 4 states may be

$$\begin{aligned}
\frac{dS_A}{dt} &= r_{sa}S_A - d_sS_A + m_sS_B - \mu_{s\to w}r_{sa}S_A + \mu_{w\to s}r_{wa}W_A \\
\frac{dW_A}{dt} &= r_{wa}W_A - d_wW_A + m_wW_B + \mu_{s\to w}r_{sa}S_A - \mu_{w\to s}r_{wa}W_A \\
\frac{dS_B}{dt} &= r_{sb}S_B + d_sS_A - m_sS_B - \mu_{s\to w}r_{sb}S_B + \mu_{w\to s}r_{wb}W_B \\
\frac{dW_B}{dt} &= r_{wb}W_B + d_wW_A - m_wW_B + \mu_{s\to w}r_{sb}S_B - \mu_{w\to s}r_{wb}W_B
\end{aligned}$$

where r, d, m, and μ show the growth rate of each state, the dispersal of a strain from the ALI to the bulk, the migration of a strain from the bulk to the ALI, and the rate of mutation, respectively.

Note that we assume reverting mutations from W to S[9,10]. This is consistent with the large mutational targets for the phenotypic reversion. For instance, many mutations in Wss operon, which is responsible for the production of the core extracellular matrix cellulose, are known to revert the phenotype from W to S. Although the combined mutation rate for reverting mutations has not been systematically measured, for the simplicity, we assumed that mutation rates to switch cell types to be equal ($\mu_{s\to w} = \mu_{w\to s}$. Also see Table S2). We confirmed that the inclusion of reverting mutations did not change the qualitative dynamics of the model, especially at the time scale of our simulations.

Here, we impose two constraints on the model to reflect the experimental setting:

1. The ALI population has a carrying capacity defined by the available space at the ALI.
2. Growth is not restricted by nutrients except for oxygen.

The second constraint is based on the fact that the growth in the bulk liquid is strongly reduced by oxygen depletion[11]. Therefore, the consumption of non-oxygen nutrients is very slow in the bulk liquid, and nutrients are continuously supplied to the ALI population by diffusion from the bulk. Considering the sufficient volume ($5 \sim 6$ mL) of a liquid medium in evolution experiments, we assume that non-oxygen nutrients are not limiting the growth throughout experiments. As the oxygen gradient can be rapidly generated (within 4 hours[11]), we implement the effect of oxygen depletion into the constant parameters for the growth rates ($r_{sa}, r_{wa} \gg r_{sb}, r_{wb}$).

With these constraints, the population size of the entire system is developed via the growth of the ALI population, whose size is upper-bounded by the spatial carrying capacity. Therefore, the entire system is supposed to grow almost linearly once the ALI population is saturated: cells are continuously produced at the ALI and transferred to the bulk liquid.

The spatial carrying capacity also means that the dispersal rate from the ALI to the bulk increases as the ALI population gets dense. We assume the crowding effect on the migration rate from the bulk to the ALI: fewer cells can move from the bulk to the ALI when the ALI population is spatially occupied. To this end, we consider the crowding-induced dispersal once the ALI population ($S_A + W_A$) reaches the spatial carrying capacity (K_A).

$$\begin{aligned}
d_sS_A &\to \frac{d_sS_A}{d_sS_A + d_wW_A}(r_{sa}S_A + r_{wa}W_A) \\
d_wW_A &\to \frac{d_wW_A}{d_sS_A + d_wW_A}(r_{sa}S_A + r_{wa}W_A)
\end{aligned}$$

In addition, we limit the migration from the bulk liquid to the ALI by

$$\begin{aligned}
m_sS_B &\to m_sS_B\left(1 - \frac{S_A + W_A}{K_A}\right) \\
m_wW_B &\to m_wW_B\left(1 - \frac{S_A + W_A}{K_A}\right)
\end{aligned}$$

Note that the growth terms are unchanged. Here, the strain competition at the ALI is mainly driven by the ratio of the dispersal d_sS_A/d_wW_A.

The parameters are set to represent the experimental situations. Namely, $r_{sa}, r_{wa} \gg r_{sb}, r_{wb}$ (oxygen depletion), $r_{sa} > r_{wa}$, $r_{sb} > r_{wb}$ (slower growth of WS than SM), $m_s > m_w$, and $d_s > d_w$ (diminished motility of WS). In terms of the strain competition at the ALI, only the $d_s > d_w$ condition (less dispersal) is regarded as the advantage for WS. In the model, we assume d_s and d_w to be constant for simplicity. If we increase d_s/d_w upon the saturation of the ALI population to represent the difference in the motile-sessile switch, evolution will be accelerated.

Supplementary Movie Captions

Supplementary Movie 1

Inoculum-dependent failure of mat formation by an adaptive Wrinkly Spreader type (WS). Mono-culture of WS could not form a mat structure when started from a low inoculum density (dilution factor 10^4). 3 mL of the cell suspension was statically cultured in a well of a 12-well plate at 28 °C. The movie was taken with a stereo microscope (Axio Zoom V16, Zeiss) in dark-field illumination with 16X magnification. The time unit is [hour:minute].

Supplementary Movie 2

Successful mat formation by WS. Mono-culture of WS could develop a robust mat when started from a high inoculum density (dilution factor 10^2). 3 mL of the cell suspension was statically cultured in a well of a 12-well plate at 28 °C. The movie was taken with a stereo microscope (Axio Zoom V16, Zeiss) in dark-field illumination with 16X magnification. The time unit is [hour:minute].

Supplementary Movie 3

Representative observation of microcolony collapse at the air-liquid interface with high magnification. WS microcolony collapse was captured at the center of a well with 140X magnification. 3 mL of WS low-inoculum culture (dilution factor 10^7) was statically cultured in a well of a 12-well plate at 28 °C. The movie was taken with a stereo microscope (Axio Zoom V16, Zeiss) in dark-field illumination.

Supplementary Movie 4

Phase separation of active fluid-like regions and sessile regions at the air-liquid interface of a co-culture of SM and WS. A mixture of SM and WS was cultured under static conditions. The movie was taken at $t = 10.5$ h for 5 seconds with bright-field imaging. The same data is also shown in Fig. S4.

Supplementary tables

Parameter	Explanation	Value
c_{max}	The maximum cell density in the bulk liquid. 10^9 [cells/mL]	5.0
r_l	The growth rate in the bulk liquid.	0.15
r	The vertical growth rate of a colony at the air-liquid interface.	0.2
m_0	The initial biomass for a newly colonized site.	1.0
K	The maximum local biomass of a colony at the air-liquid interface.	80
G	The coefficient for the stability of a colony.	300
R_{end}	The ratio of colonized sites of the air-liquid interface to end a simulation.	0.85
C_μ	The coefficient to relate $c(t)$ and the migration rate μ.	10^{-5}
L_{box}	The system size of the air-liquid interface. $L_{box} \times L_{box}$ was the entire 2D space.	500
r_s	The rate of swimming events.	0 or 0.05
σ	The spreading of the swimming distance.	5.0
T_{mass}	The minimum time step for the mass growth and the stability evaluation.	0.1
ε	The relative matrix production of the motile non-biofilm former.	0.1

Table S1. List of parameters in the spatial model simulations.

Parameter	Explanation	Value
r_{sa}	The growth rate of SM at the ALI.	1.0
r_{sb}	The growth rate of SM in the bulk liquid.	0.001 r_{sa}
r_{wa}	The growth rate of WS at the ALI.	0.85
r_{wb}	The growth rate of WS in the bulk liquid.	0.001 r_{wa}
d_s	The dispersal rate of SM from the ALI to the bulk liquid.	0.002
d_w	The dispersal rate of WS from the ALI to the bulk liquid	0.001
m_s	The migration rate of SM from the bulk liquid to the ALI	0.02
m_w	The migration rate of WS from the bulk liquid to the ALI	0.01
$\mu_{s\to w}$	The rate of mutation from SM to WS.	1.1×10^{-8}[12]
$\mu_{w\to s}$	The rate of mutation from WS to SM.	1.1×10^{-8}
K_A	The spatial carrying capacity of the ALI population.	10^{10}
S_{A0}	The initial population size of SM at the ALI.	10^5
W_{A0}	The initial population size of WS at the ALI.	0
S_{b0}	The initial population size of SM in the bulk liquid.	10^7
W_{b0}	The initial population size of WS in the bulk liquid.	0

Table S2. List of parameters in the four-state model simulations.

Strain	Mat	Genotype	Internal ID	Reference
SBW25 wild type (SM)	No		MPB13805 (PBR341)	[13–15]
Wrinkly spreader (WS) *wspF*	Yes	*wspF* (PFLU1224) Δ150bp (677-826).	MPB08384	[16]
Wrinkly spreader (WS) *awsX*	Yes	*awsX* (PFLU5211) Δ32bp (229-261).	MPB10583	[17]
Wrinkly spreader (WS) *mwsR*	Yes	*mwsR* (PFLU5329) T2183C.	MPB10613	[17]
WS(*wspF*)-Δ*fliA*	Yes*	*wspF* (PFLU1224) Δ150bp (677-826), Δ*fliA* (PFLU4417).	MPB36210	This work
SM-Δ*fliA*	No	Δ*fliA* (PFLU4417).	MPB14427	This work
SM-Δ*wss*	No	Δ*wss* (PFLU0300-0309).	MPB14424	This work
WS(*wspF*)-mScarlet	Yes	*wspF* (PFLU1224) Δ150bp (677-826), attTn7::Tn7-PnptII-mScarlet-I-Gm.	MPB16290	This work†
WS(*wspF*)-Δ*pvdS*-sGFP2	Yes	*wspF* (PFLU1224) Δ150bp (677-826), Δ*pvdS* (PFLU4389), attTn7::Tn7-PnptII-sGFP2-Kan.	MPB23958	This work‡
WS(*wspF*)-Δ*pvdS*-mScarlet	Yes	*wspF* (PFLU1224) Δ150bp (677-826), Δ*pvdS* (PFLU4389), attTn7::Tn7-PnptII-mScarlet-I-Gm.	MPB23959	This work‡
SM-mScarlet	No	attTn7::Tn7-PnptII-mScarlet-I-Gm.	MPB15845	[18]
SM-Δ*pvdS*-sGFP2	No	Δ*pvdS* (PFLU4389), attTn7::Tn7-PnptII-sGFP2-Kan.	MPB26812	This work‡
SM-Δ*pvdS*-mScarlet	No	Δ*pvdS* (PFLU4389), attTn7::Tn7-PnptII-mScarlet-I-Gm.	MPB26813	This work‡
SM pCdrA-mScarlet	No	pCdrA-mScarlet (modified from Addgene 111614): plasmid cdrA-RBSII-mScarlet-T0-T1, pRO1600, oriV; GmR and AmR[19]	MPB38082	This work

Table S3. The list of *Pseudomonas fluorescens* SBW25 strains used in this paper. The strains without references were obtained from the Microbial Population Biology strain repository (Max Planck Institute for Evolutionary Biology, Germany), and they were constructed by a two-step allelic exchange protocol. *wspF*, *awsX*, and *mwsR* are negative regulators of c-di-GMP[17], and mutations on the loci typically lead to the overexpression of c-di-GMP. Deleting *wss* disables the production of cellulose, the main extracellular matrix of *P. fluorescens* SBW25. Deleting *fliA* makes cells immotile, and deleting *pvds* eliminates the expression of the fluorescent siderophore pyoverdin. * A mat is successfully formed only from a very high inoculum ($\sim 10^8$ cells/mL). † mScarlet is integrated from pMRE-Tn7-145 (Addgene 118561), and sGFP2 is integrated from pMRE-Tn7-152 (Addgene 118566)[20]. ‡ mScarlet and sGFP2 are integrated in the Δ*pvdS* strains in the previous work[21].

Supplementary figures

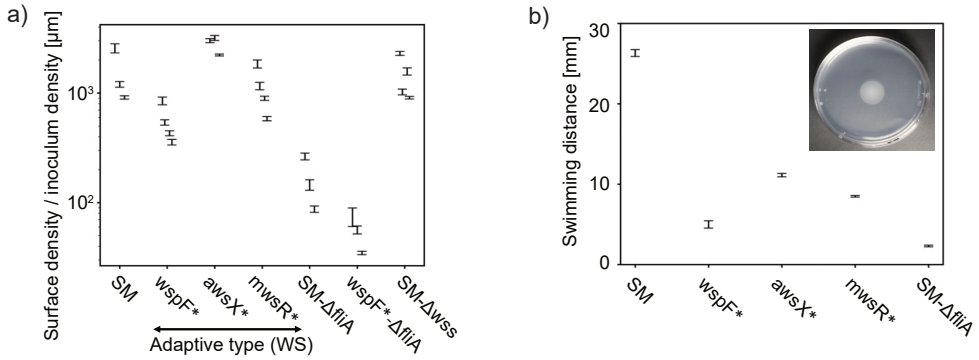


Figure S1. Measurements of motility and early ALI occupation with various strains. (a) The efficiency of ALI occupation was quantified by the ratio of the surface density at $t = 2$ h over the inoculum volume density. The surface density was measured by counting the number of cells at the ALI with microscopy. 3 or 4 inoculum densities were tested for each strain, shown as different points in the plot (from left to right, the inoculum densities are low to high). (b) Motility assay on semi-solid agar plates. The inset shows the representative motility assay on a 10 cm semi-solid agar plate.

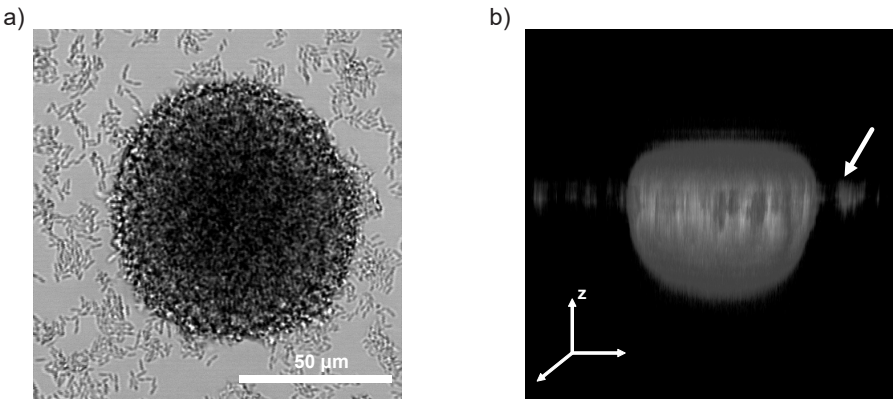


Figure S2. 3D structure of a WS microcolony at an ALI. (a) A top view of a microcolony at an air-liquid interface. (b) The 3D reconstruction of the microcolony in (a) by a confocal microscope. The z-position of small cell clusters (white arrow) was lower than the top surface of the microcolony, suggesting that the microcolony penetrated the air-liquid interface.

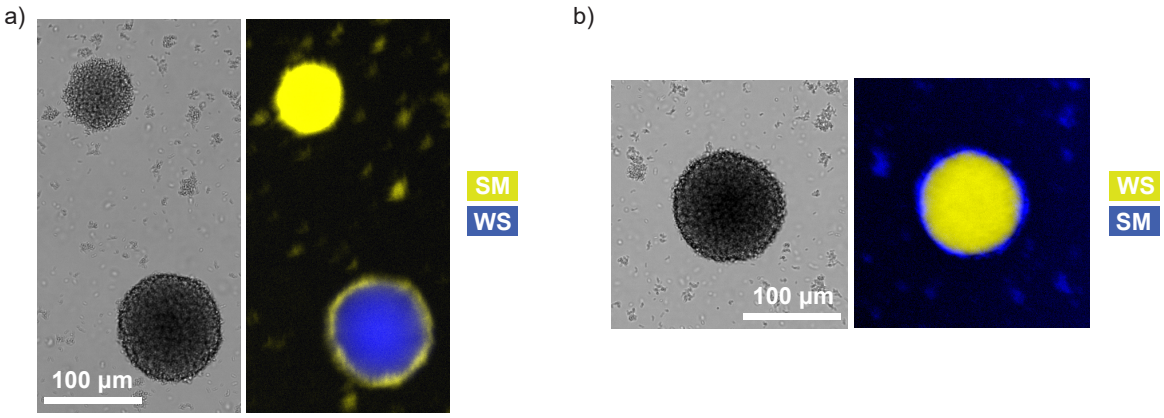


Figure S3. Adhesion of the wild type strain (SM) to microcolonies of the mat-forming strain (WS). The SM and WS strains were co-cultured starting from a low inoculum ($\sim 5 \times 10^3$ cells/mL for each), and imaged by fluorescent microscopy at $t = 16$ h. The flipped combinations of the fluorescence proteins were shown in (a) (SM-mScarlet, WS-sGFP2) and (b) (SM-sGFP2, WS-mScarlet).

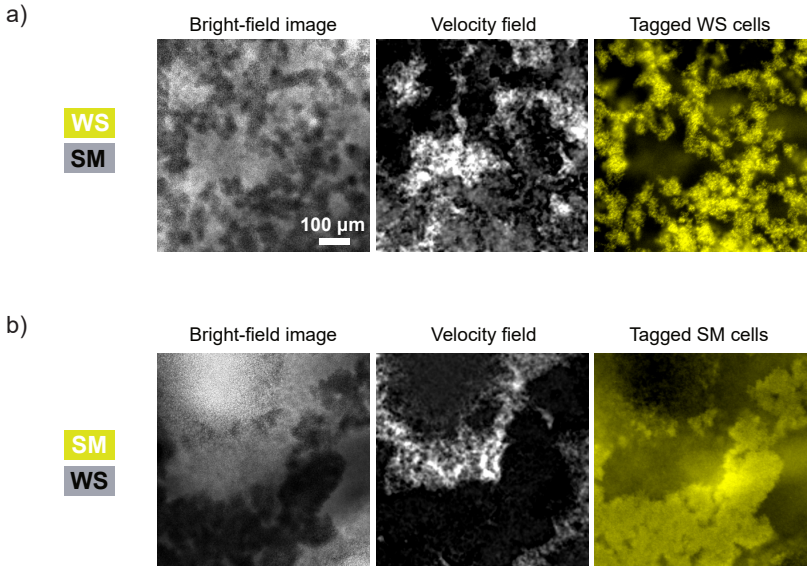


Figure S4. Inclusion of SM in the sessile regions during transition to motility at the ALI. Snapshots and PIV velocity fields were shown for co-cultures of SM and WS-mScarlet (a) and SM-mScarlet and WS (b). For the velocity field, the absolute value of the velocity at each position was averaged over 5 seconds. The images were taken at $t = 10.5$ h after inoculation.

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